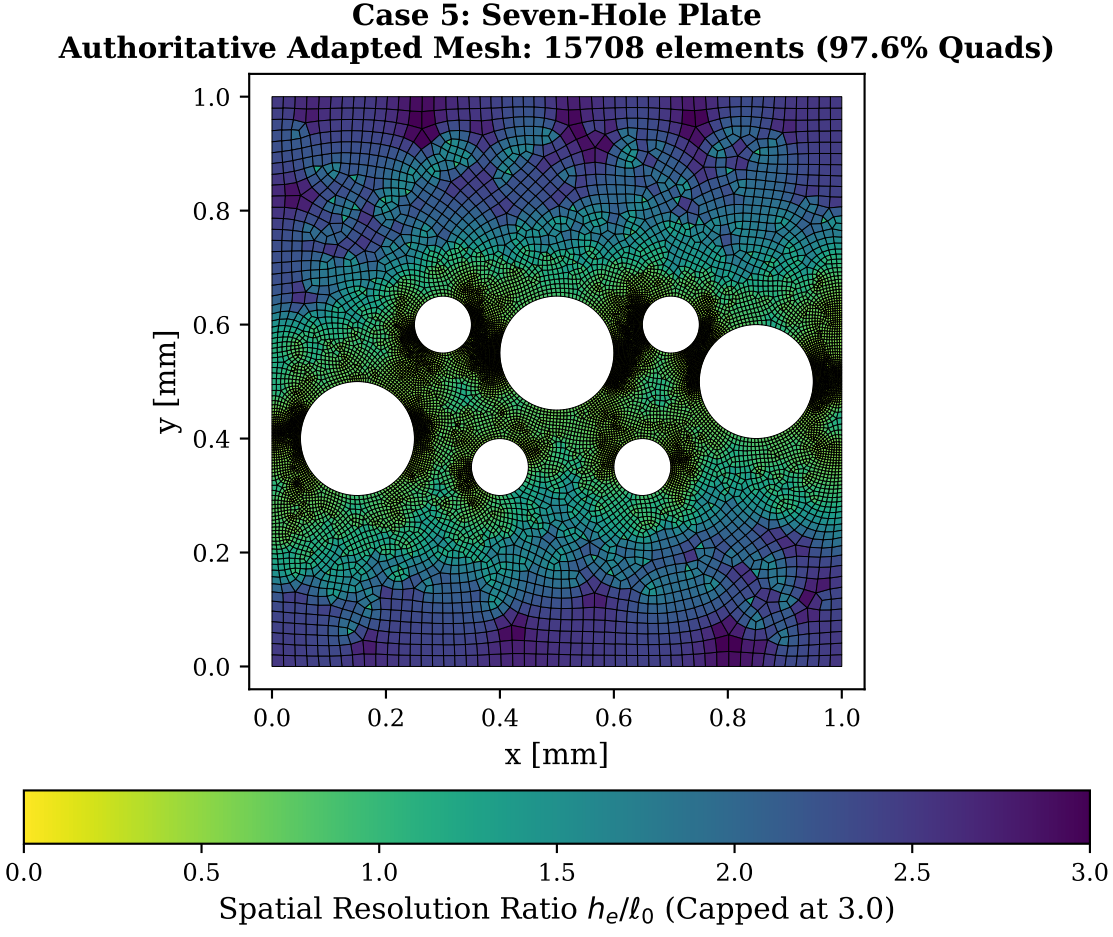
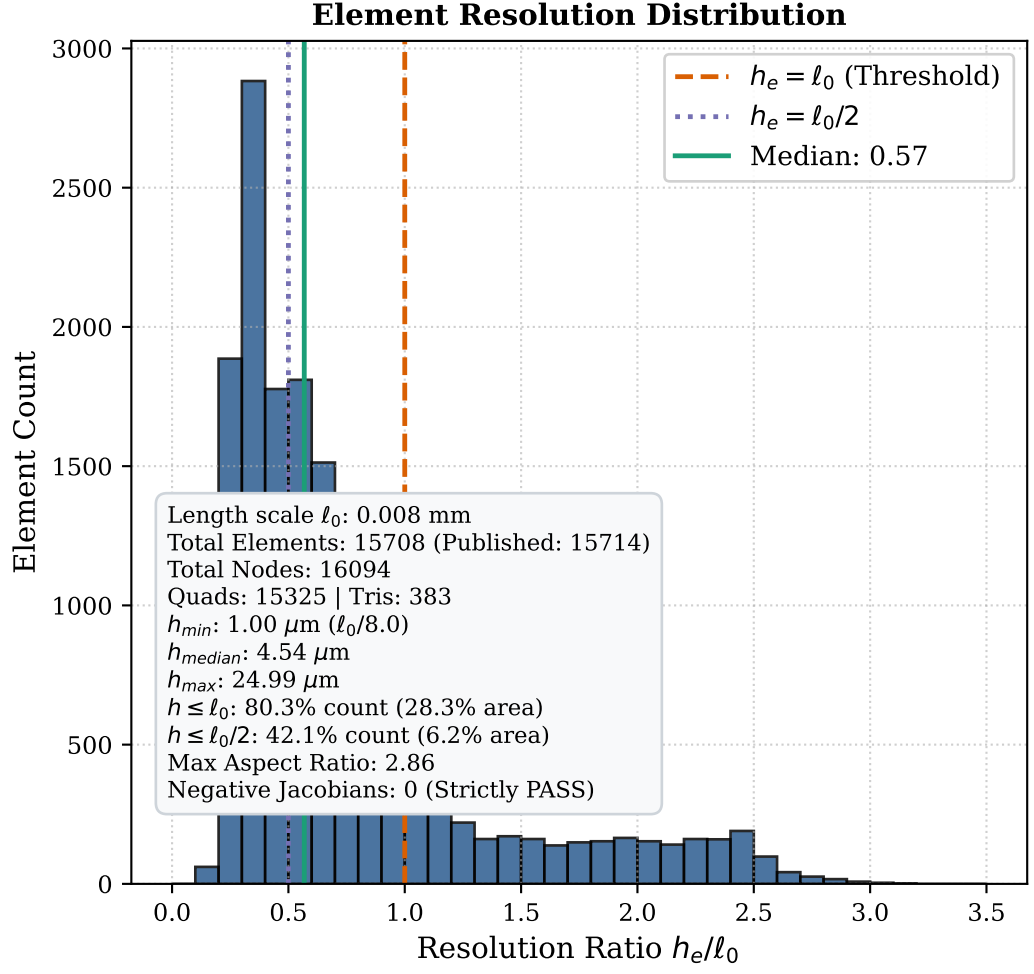

Case 5: Seven-Hole Plate
Authoritative Adapted Mesh: 15708 elements (97.6% Quads)
y [mm]
x [mm]
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Spatial Resolution Ratio h_e/ℓ_0 (Capped at 3.0)


Element Resolution Distribution
$h_e = \ell_0$ (Threshold)
$h_e = \ell_0/2$
Median: 0.57
Length scale ℓ_0: 0.008 mm
Total Elements: 15708 (Published: 15714)
Total Nodes: 16094
Quads: 15325 | Tris: 383
h_{min}: 1.00 μm (ℓ_0/8.0)
h_{median}: 4.54 μm
h_{max}: 24.99 μm
$h \le \ell_0$: 80.3% count (28.3% area)
$h \le \ell_0/2$: 42.1% count (6.2% area)
Max Aspect Ratio: 2.86
Negative Jacobians: 0 (Strictly PASS)
Element Count
Resolution Ratio h_e/ℓ_0
0
500
1000
1500
2000
2500
3000
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5